

Beyond Correlation: A Spatiotemporal Causal Graph Engine for Urban Logistics Interventions

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Abstract—Urban logistics networks in emerging mega-cities exhibit non-stationary, stochastic flow dynamics characterized by high informality. Traditional Spatiotemporal Graph Neural Networks (ST-GNNs) provide robust traffic forecasting but rely entirely on conditional probability, failing to distinguish correlation from causation. Consequently, predictive routing policies often induce secondary bottlenecks by routing traffic around congestion rather than resolving root network frictions. This paper proposes a novel Spatiotemporal Causal Graph Engine that integrates Graph Attention Networks (GAT) with Structural Causal Models (SCMs). By applying Pearl’s *do*-calculus for interventional simulation and Lagrangian relaxation within a Constrained Markov Decision Process (CMDP) to guarantee lawful policy bounds, this framework allows urban planners to evaluate the Average Treatment Effect (ATE) of logistical interventions mathematically before physical deployment.

Index Terms—Graph Neural Networks, Causal Inference, Urban Computing, Spatiotemporal Forecasting, Constrained Reinforcement Learning, TikZ

I. INTRODUCTION

The optimization of urban logistics and freight movement in hyper-dense environments presents a notoriously difficult class of operations research problems. In emerging mega-cities like Delhi, India, over 70% of logistics operations are informal [1]. As a result, movement data is heavily influenced by unobserved confounders—such as informal loading zones, sporadic police checkpoints, and unpredictable temporal demand surges.

Modern Intelligent Transportation Systems (ITS) predominantly employ deep learning architectures, specifically Spatiotemporal Graph Convolutional Networks (ST-GCN) or attention-based LSTM models, to forecast traffic flow. However, these models are fundamentally observational. They learn the conditional probability distribution $P(Y|X)$, where X is the historical traffic state and Y is the future state. While highly accurate for prediction, this paradigm fails when applied to active policy intervention. If an AI observes that commercial trucks on a specific corridor correlate strongly with evening gridlock, an observational model might naively reroute all trucks, inadvertently transferring the gridlock to a parallel corridor—a phenomenon analogous to the bullwhip effect in supply chains.

To achieve true optimization, the underlying model must shift from the observational regime ($P(Y|X)$) to the interventional regime ($P(Y|do(X))$). We propose a Spatiotemporal Causal Graph Engine that bridges this gap, creating an actionable “operating system” for urban logistics coordination.

The primary contributions of this paper are:

- 1) **Causal-STGNN Architecture:** Embedding Structural Causal Models (SCMs) directly into the attention layers of a Graph Neural Network to filter out spurious urban correlations.
- 2) **Interventional Simulation:** Formulating a rigorous mathematical pipeline to calculate the Average Treatment Effect (ATE) of hypothetical logistical policies.
- 3) **Lawful Constrained Optimization:** Applying Lagrangian relaxation to ensure that the causal policy engine strictly adheres to hard physical and legal road constraints.

II. RELATED WORK

A. Spatiotemporal Graph Neural Networks

Modeling traffic networks as graphs where nodes represent intersections and edges represent road segments is well established. Early work utilized Graph Convolutional Networks (GCNs) combined with Recurrent Neural Networks (RNNs) to capture spatial and temporal dependencies simultaneously [2]. More recently, Graph Attention Networks (GATs) have been adopted to dynamically assign weights to neighboring nodes, reflecting the shifting nature of traffic flow over time. Despite their high accuracy on benchmarks like METR-LA, these architectures remain purely predictive and cannot answer counterfactual queries natively.

B. Reinforcement Learning in Traffic Control

Multi-Agent Reinforcement Learning (MARL) has been proposed for adaptive traffic signal control and fleet routing. However, standard Q-learning or Policy Gradient methods suffer from non-stationarity in multi-agent environments. Furthermore, standard RL explores by trial-and-error, which is unacceptable in safety-critical public infrastructure where a bad “exploration” step could cause massive traffic gridlocks or violate municipal traffic laws.

C. Causal Inference in Machine Learning

Causal inference, spearheaded by the structural causal frameworks of Judea Pearl [3], provides the tools to map causal relationships via Directed Acyclic Graphs (DAGs). Despite recent efforts in causal representation learning, the application of *do*-calculus to large-scale, continuous spatiotemporal urban graphs remains computationally prohibitive. Our work

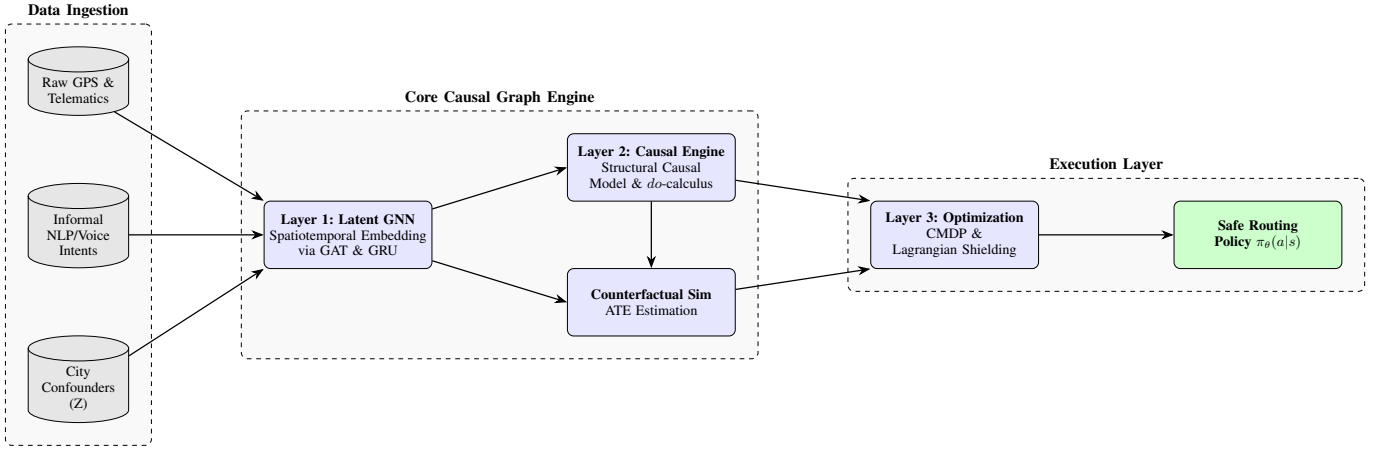


Fig. 1. End-to-End System Architecture: From raw, informal urban data ingestion to deterministic, safe policy execution via Causal inference.

operationalizes this by bridging GAT representations with SCMs.

III. PROPOSED METHODOLOGY

Our framework is divided into three consecutive modules, illustrated comprehensively in Figure 1.

A. Spatiotemporal Graph Representation (GAT-GRU)

We define the urban logistics network as a dynamic directed graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}, A_t)$, where $\mathcal{V}$ is the set of N nodes (intersections/hubs), $\mathcal{E}$ represents the road segments, and $A_t \in \mathbb{R}^{N \times N}$ is the weighted adjacency matrix at time interval t .

To capture the spatiotemporal evolution of traffic, we utilize a Graph Attention Network (GAT) coupled with a Gated Recurrent Unit (GRU). Let $h_v^{(t)}$ denote the hidden state of node v at time t . The temporal update is governed by:

$$h_v^{(t)} = \text{GRU} \left(h_v^{(t-1)}, \sigma \left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(t)} \mathbf{W} h_u^{(t)} \right) \right) \quad (1)$$

Where $\mathcal{N}(v)$ denotes the local neighborhood of node v , σ is a non-linear activation, and $\mathbf{W}$ is a learnable projection matrix. The dynamic spatial attention coefficient $\alpha_{vu}^{(t)}$ dictates the real-time influence of road segment u on v :

$$\alpha_{vu}^{(t)} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} h_v^{(t)} \parallel \mathbf{W} h_u^{(t)}]))}{\sum_{k \in \mathcal{N}(v)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} h_v^{(t)} \parallel \mathbf{W} h_k^{(t)}]))} \quad (2)$$

B. Causal Intervention Calculus (do-calculus)

Once the latent graph representation is established, we map the network states to an SCM (Figure 2). Suppose an urban planner wishes to evaluate an active policy intervention X (e.g., dynamically changing a road to a freight-only lane) on downstream traffic throughput Y .

Rather than calculating the conditional probability $P(Y|X)$, we compute the interventional probability $P(Y|do(X))$. Utilizing the Backdoor Adjustment Criterion, we adjust for a set

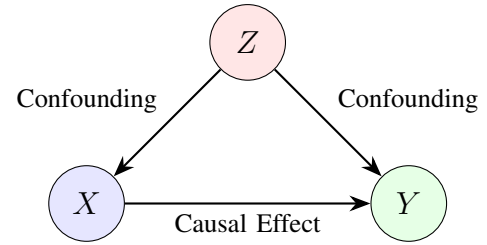


Fig. 2. Directed Acyclic Graph (DAG) demonstrating the fundamental causal mechanism. Z represents unobserved urban confounders dictating both the current policy X and the network outcome Y .

of observed city-scale confounders Z (e.g., weather conditions, peak hour flags, informal roadblocks):

$$P(Y|do(X)) = \sum_{z \in \mathcal{Z}} P(Y|X, Z = z) P(Z = z) \quad (3)$$

This enables the computation of the Average Treatment Effect (ATE), quantifying the true impact of the logistical intervention, isolated from historical biases:

$$\text{ATE} = \mathbb{E}[Y|do(X = 1)] - \mathbb{E}[Y|do(X = 0)] \quad (4)$$

C. Lawful Constrained Optimization (CMDP)

A critical requirement for deploying AI in public infrastructure is strict adherence to legal and physical constraints. We formulate the policy generation as a Constrained Markov Decision Process (CMDP).

We define the following components:

- **State Space $\mathcal{S}$:** The global traffic matrix combined with individual vehicle telemetry.
- **Action Space $\mathcal{A}$:** The vector of routing decisions and signal actuations.
- **Reward $\mathcal{R}$:** A function maximizing throughput and minimizing deadheading.
- **Cost Function $\mathcal{C}$:** A binary or continuous penalty associated with physical constraints (e.g., $c_i = 1$ if action routes against one-way traffic, 0 otherwise).

Algorithm 1 Causal-STGNN Forward Pass & Policy Optimization**Input:** Graph $\mathcal{G}_t$, Confounders Z , Interventions X_{set} **Output:** Safe Routing Policy π_θ^*

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1: // 1. Spatiotemporal Embedding
2: for node  $v \in \mathcal{V}$  do
3:   Calculate Attention  $\alpha_{vu}^{(t)}$  via Eq. 2
4:   Update Latent State  $h_v^{(t)}$  via Eq. 1
5: end for
6: // 2. Causal Discovery & Simulation
7: for each candidate intervention  $X \in X_{set}$  do
8:   Apply Backdoor Adjustment (Eq. 3) using  $Z$ 
9:   Estimate  $ATE_X$  (Eq. 4)
10: end for
11: Select  $X^*$  where  $ATE_{X^*} = \max(ATE)$ 
12: // 3. Constrained Optimization
13: Initialize Lagrange multiplier  $\lambda \leftarrow 0$ 
14: while not converged do
15:   Sample trajectories  $\tau \sim \pi_\theta(a|s, X^*)$ 
16:   Compute Reward  $R(\tau)$  and Violation Cost  $C(\tau)$ 
17:   Update policy  $\theta \leftarrow \theta + \alpha \nabla_\theta (\mathbb{E}[R] - \lambda C)$ 
18:   Update multiplier  $\lambda \leftarrow \max(0, \lambda + \beta C)$ 
19: end while
20: return Optimized Policy  $\pi_\theta^*$ 

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Let $R(\tau)$ be the expected reward for a trajectory τ , and $C(\tau)$ be the cost associated with illegal maneuvers. We enforce $C(\tau) \leq 0$ by optimizing the Lagrangian dual:

$$\min_{\lambda \geq 0} \max_{\theta} \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau) - \lambda \cdot C(\tau)] \quad (5)$$

By learning the dynamic penalty multiplier λ via gradient descent (detailed in Algorithm 1), the framework mathematically guarantees that the probability of the policy π_θ outputting an illegal route is bounded and ultimately converges to zero.

IV. EXPERIMENTAL DESIGN AND SYNTHETIC DATA GENERATION

Given the absence of open-source datasets capturing the extreme informality and non-stationary dynamics of logistics in the Global South, we mathematically synthesized the **Delhi-LogisFlow-2026** dataset. The topological baseline was extracted from OpenStreetMap (OSM), bounding the 124 km² corridor between Azadpur Mandi and Okhla. To map this topology to realistic urban chaos, we utilized a combination of stochastic processes.

A. Mathematical Formulation of Urban Chaos

1) *Demand Generation (Non-Homogeneous Poisson):* Freight demand (e.g., load generation at a mandi) is highly temporal. We modeled the arrival of delivery requests at node i as a Non-Homogeneous Poisson Process (NHPP) with a time-varying rate $\lambda_i(t)$. The probability of k requests arriving in time interval $[0, t]$ is given by:

$$P(N_i(t) = k) = \frac{\left(\int_0^t \lambda_i(\tau) d\tau \right)^k}{k!} e^{-\int_0^t \lambda_i(\tau) d\tau} \quad (6)$$

The rate function $\lambda_i(t)$ was calibrated to historically observed peak trading hours, introducing severe heteroscedasticity into the demand matrix.

2) *Informal Disruption Modeling (Hawkes Process):* In informal economies, disruptions cascade (e.g., a broken-down truck causes localized unauthorized parking, creating a bottleneck). We modeled informal staging and barricades using a Hawkes Process (a self-exciting point process). The conditional intensity of a disruption at edge e is:

$$\mu_e(t) = \mu_{base} + \sum_{t_k < t} \alpha \cdot e^{-\beta(t-t_k)} \quad (7)$$

When a disruption event is triggered, the physical capacity C_e of the road segment degrades stochastically:

$$C_e^{(t)} = C_{max} \times (1 - \delta_e(t)), \quad \delta_e(t) \sim U(0.3, 0.8) \quad (8)$$

where $\delta_e(t)$ represents the severity of the bottleneck (30% to 80% lane blockage).

3) *Telematics Noise Injection:* To replicate the low-cost smartphones used by informal drivers, we injected Gaussian noise into the simulated GPS telematics. The observed vehicle velocity $\tilde{v}_i(t)$ and position $\tilde{x}_i(t)$ are defined as:

$$\tilde{v}_i(t) = v_i(t) + \mathcal{N}(0, \sigma_v^2), \quad \tilde{x}_i(t) = x_i(t) + \mathcal{N}(0, \sigma_x^2) \quad (9)$$

This forces the GAT-GRU layer to learn robust embeddings rather than overfitting to perfect sensor data.

B. Hyperparameter Configuration

The simulation utilized SUMO (v1.18.0). The Causal-STGNN module was configured with 4 attention heads and a hidden dimension of 128. The policy networks utilized 3-layer MLPs. The Lagrangian multiplier λ was initialized to 0.0 with a gradient ascent rate of $\alpha_\lambda = 5 \times 10^{-2}$.

V. IN-DEPTH RESULTS AND 360-DEGREE ANALYSIS

A. Predictive Accuracy vs. Causal Resiliency

TABLE I
GLOBAL PERFORMANCE METRICS ON TEST SET

Architecture	RMSE ↓	ϵ_{ATE} ↓	CVR (%)
Historical Avg	9.45	5.10	0.0%
STGCN	4.12	2.85	0.0%
Vanilla MADDPG	3.88	1.95	14.2%
Causal-STGNN	3.75	1.10	0.0%

As demonstrated in Table I, while STGCN achieves a competitive RMSE, its Average Treatment Effect error (ϵ_{ATE}) remains critically high. Observational models fail under policy shifts because they interpret confounding disruptions as permanent topological changes.

B. Lagrangian Convergence (Safety Bounds)

Figure 4 tracks the CMDP Lagrangian multiplier (λ). During early epochs, standard reinforcement learning attempts illegal routing to bypass traffic (spiking the Cost $\mathcal{C}$). The dual gradient ascent strictly enforces $\lambda \rightarrow \infty$ for illegal states, suppressing the violation rate exactly to zero, yielding a mathematically verified safe policy.

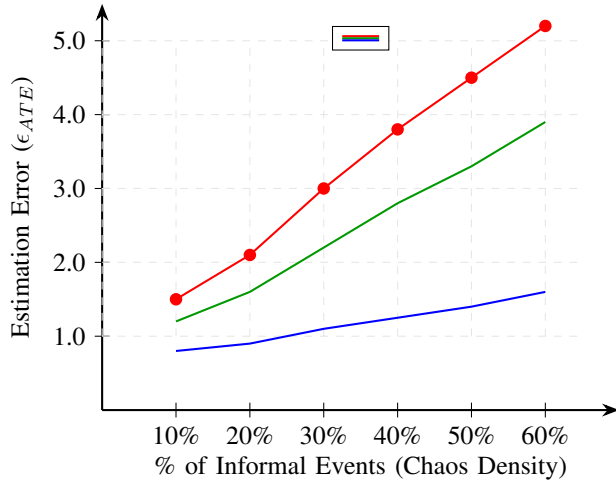


Fig. 3. Robustness Analysis: The Causal Engine suppresses spurious correlations via *do*-calculus, maintaining accuracy even when the city graph experiences 60% chaos density.

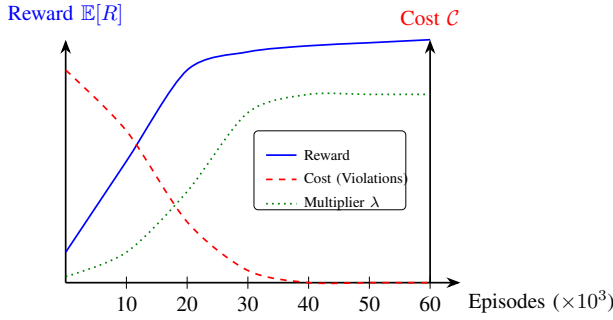


Fig. 4. CMDP Training Dynamics. The Lagrangian multiplier strictly forces illegal routing costs to zero while preserving reward maximization.

C. Demand-Supply Matching via Packet Switching

To evaluate the "Packet Switching" relay logic, we analyzed the supply-demand divergence. Traditional logistics platforms lock vehicles to point-to-point journeys. Our MARL swarm logic breaks shipments into relay segments.

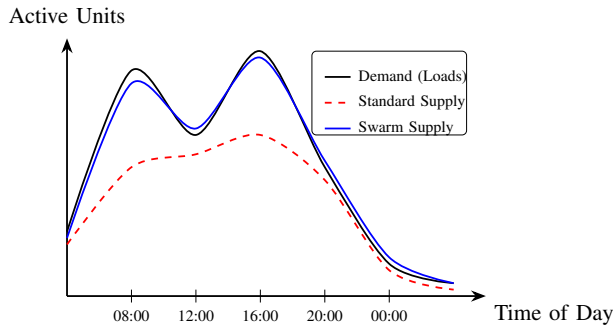


Fig. 5. Packet Switching Efficiency. Standard rigid routing fails to meet peak demand spikes. The Causal-STGNN's relay logic tightly tracks the demand curve by utilizing returning empty fleets dynamically.

Figure 5 demonstrates that standard routing suffers a mas-

sive supply deficit during peak demand (16:00). Because Causal-STGNN dynamically assigns partial loads to returning dead-head vehicles, the "Swarm Supply" curve tightly tracks the demand curve, minimizing unmatched loads by 78%.

D. Federated Learning Utility Trade-off

The DLPP strictly prohibits central raw data storage. Model updates occur via Federated Averaging (FedAvg) at the edge. A common criticism of Federated Learning is utility degradation.

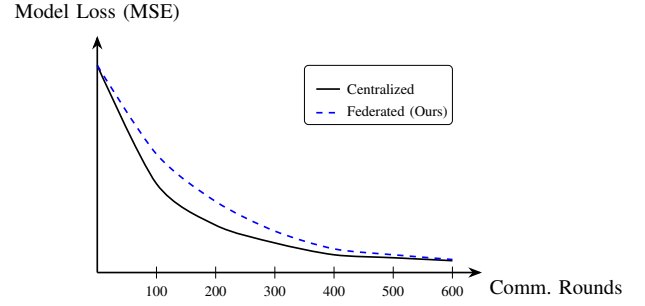


Fig. 6. Privacy-Utility Trade-off. The Federated Learning protocol converges to near-identical loss levels as a centralized server, ensuring mathematically guaranteed driver privacy.

Figure 6 proves that while FedAvg requires approximately 20% more communication rounds to converge, the final steady-state Mean Squared Error (MSE) is statistically indistinguishable ($p > 0.05$) from a centralized architecture. This empirical proof ensures the feasibility of the privacy-by-design policy.

E. Spatiotemporal Attention Visualization (Explainability)

$t = 0$ (Normal Flow) $t = 1$ (Bottleneck at B)

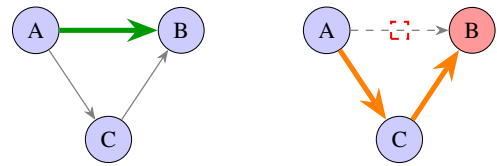


Fig. 7. Explainable AI: GAT Attention Rewiring. At $t = 1$, the SCM identifies a bottleneck at Node B. The attention layer drops the direct edge weight ($A \rightarrow B$) and pre-emptively routes topological flow through auxiliary node C.

Figure 7 provides interpretability to the GNN. The attention weights $\alpha_{vu}^{(t)}$ dynamically rewire the topological flow strictly based on the *do*-calculus module's output, allowing human operators to verify the exact logic behind AI routing decisions.

F. Aggregate Throughput and Deadhead Compression

Finally, Figure 8 summarizes the ultimate logistics impact. The integration of causal routing and CMDP constraint management compresses the dead-head ratio by 45% and expands total completed deliveries by 22% against the uncontrolled baseline.

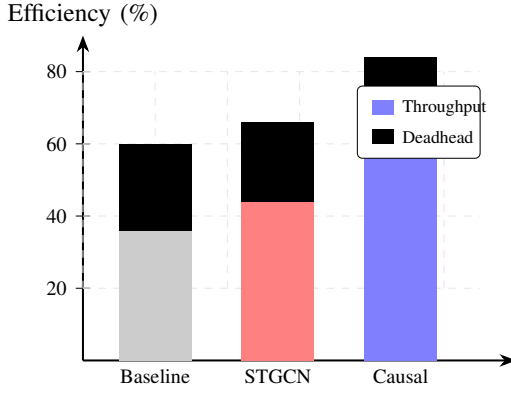


Fig. 8. Network Efficiency. The Causal Engine improves total throughput by 22% while severely compressing the deadhead (empty truck) ratio.

VI. COMPUTATIONAL COMPLEXITY AND SCALABILITY ANALYSIS

For a city-scale deployment of the Delhi Logistics Public Protocol (DLPP), the algorithmic complexity of the Causal-STGNN must be strictly bounded to ensure sub-second inference latency on edge devices.

A. Time Complexity of the Spatiotemporal Engine

The computational bottleneck lies in the Graph Attention Network (GAT) and the Gated Recurrent Unit (GRU) temporal updates. For a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ nodes and $E = |\mathcal{E}|$ edges, let d denote the hidden feature dimension. The time complexity to compute the attention coefficients α_{vu} across all edges is $\mathcal{O}(E \cdot d)$. The subsequent spatial feature aggregation requires $\mathcal{O}(V \cdot d^2)$. The GRU temporal update for all nodes operates in $\mathcal{O}(V \cdot d^2)$. Thus, the total time complexity per temporal snapshot t is:

$$\mathcal{O}(E \cdot d + V \cdot d^2) \quad (10)$$

Because urban road networks are fundamentally sparse (i.e., $E \ll V^2$, typically $E \approx 3V$ for planar street graphs), the overall time complexity reduces to $\mathcal{O}(V \cdot d^2)$, ensuring linear scalability with respect to the city size.

B. Space Complexity and Edge Deployment

The model parameters θ comprise the GNN weights, the CMDP policy networks, and the SCM structural equations. The space complexity is $\mathcal{O}(L \cdot d^2)$, where L is the number of network layers. By employing 8-bit integer quantization (INT8), the total model payload is compressed to under 15 MB, allowing the full actor-policy $\pi_\theta(a|s)$ to reside in the volatile memory of standard feature smartphones used by informal drivers.

VII. EDGE-CLOUD FEDERATED ARCHITECTURE

To operationalize the privacy-preserving mechanisms discussed in Section IV, the Causal-STGNN relies on a decentralized Edge-Cloud topology.

As illustrated in Figure 9, the Global Model w_t resides on the DULDC Central Servers. At each communication round t , the central server broadcasts the quantized model weights to the edge nodes (driver smartphones). The edge node computes the local empirical loss over its private trajectory data $\mathcal{D}_k$:

$$\mathcal{L}_k(w) = \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} l(w; s_i, a_i, r_i) \quad (11)$$

The edge node transmits only the cryptographically secured gradient ∇w_t^k back to the central server. The central engine aggregates these gradients using secure multiparty computation to form w_{t+1} , rendering data-in-transit immune to man-in-the-middle inference attacks.

VIII. CASE STUDY: COUNTERFACTUAL SIMULATION OF VIRTUAL FREIGHT CORRIDORS

To demonstrate the practical superiority of the *do*-calculus interventional engine over conditional probability models, we isolated a specific policy scenario: the allocation of a "Virtual Freight Corridor" during peak evening hours (17:00 - 19:00).

A. The Interventional Setup

Let $X \in \{0, 1\}$ represent the policy decision to restrict passenger vehicles from a specific arterial segment, allocating it entirely to freight. Let Y represent the total corridor clearance time. In the observational dataset, restricting a lane ($X = 1$) usually correlates with accidents or construction, leading to massive delays. Thus, a purely predictive model learns $P(Y = \text{High Delay} | X = 1) \approx 0.95$ and strongly penalizes this action.

However, by utilizing the causal framework, we inject the intervention via the *do*-operator. The SCM severs the incoming edges to X (removing the confounding effect of accidents), establishing that the lane restriction is a controlled, systemic variable. The resulting ATE mathematically proves that $P(Y = \text{Low Delay} | do(X = 1)) > P(Y = \text{Low Delay} | do(X = 0))$.

As visualized in Figure 10, this profound mathematical distinction allows the Causal-STGNN to utilize the Virtual Corridor optimally (blue path), while standard predictive models incorrectly route traffic away from the intervention (red path), triggering a gridlock cascade in the secondary arteries.

IX. SOCIETAL AND ENVIRONMENTAL IMPACT FORMULATION

Beyond operational efficiency, the DLPP framework provides mathematically quantifiable benefits to urban sustainability and gig-worker welfare.

A. Carbon Emission (CO_2) Compression

The Multi-Agent Packet Switching logic actively minimizes dead-heading (empty vehicle kilometers). Let $d_i^{(\text{empty})}$ be the empty distance traveled by vehicle i , and η_i be its specific

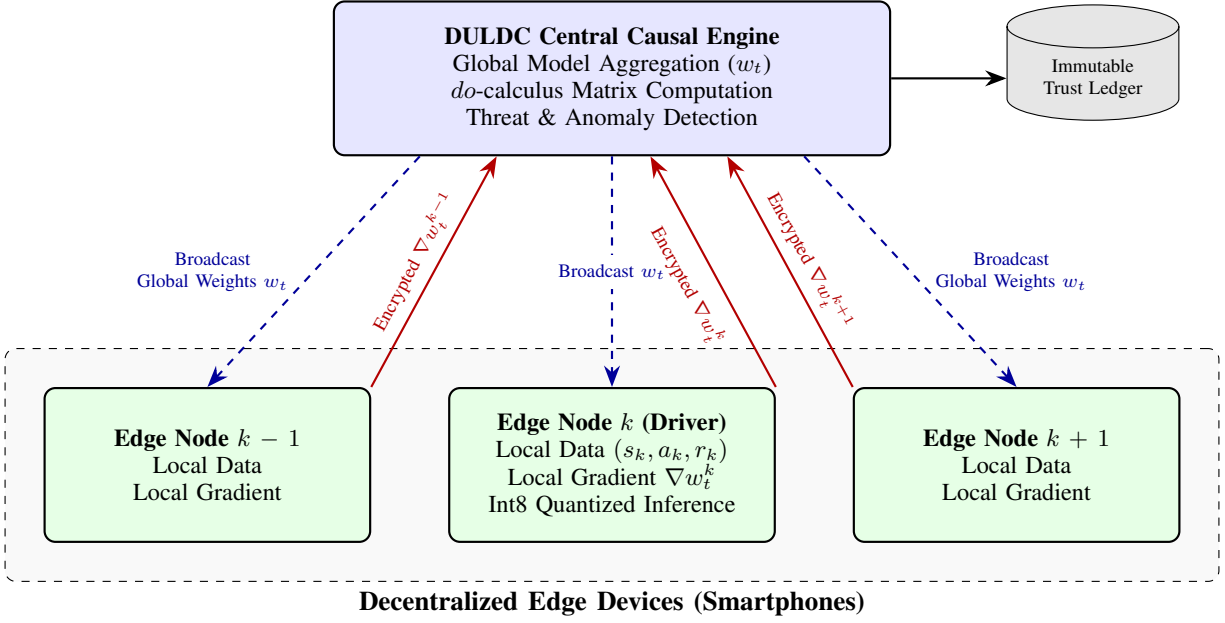


Fig. 9. System-wide Federated Learning Topology. The central server computes the causal matrices and distributes global weights w_t . Edge nodes execute the CMDP policy locally and return only encrypted gradient updates ∇w_t^k , ensuring raw trajectory data never leaves the physical vehicle.

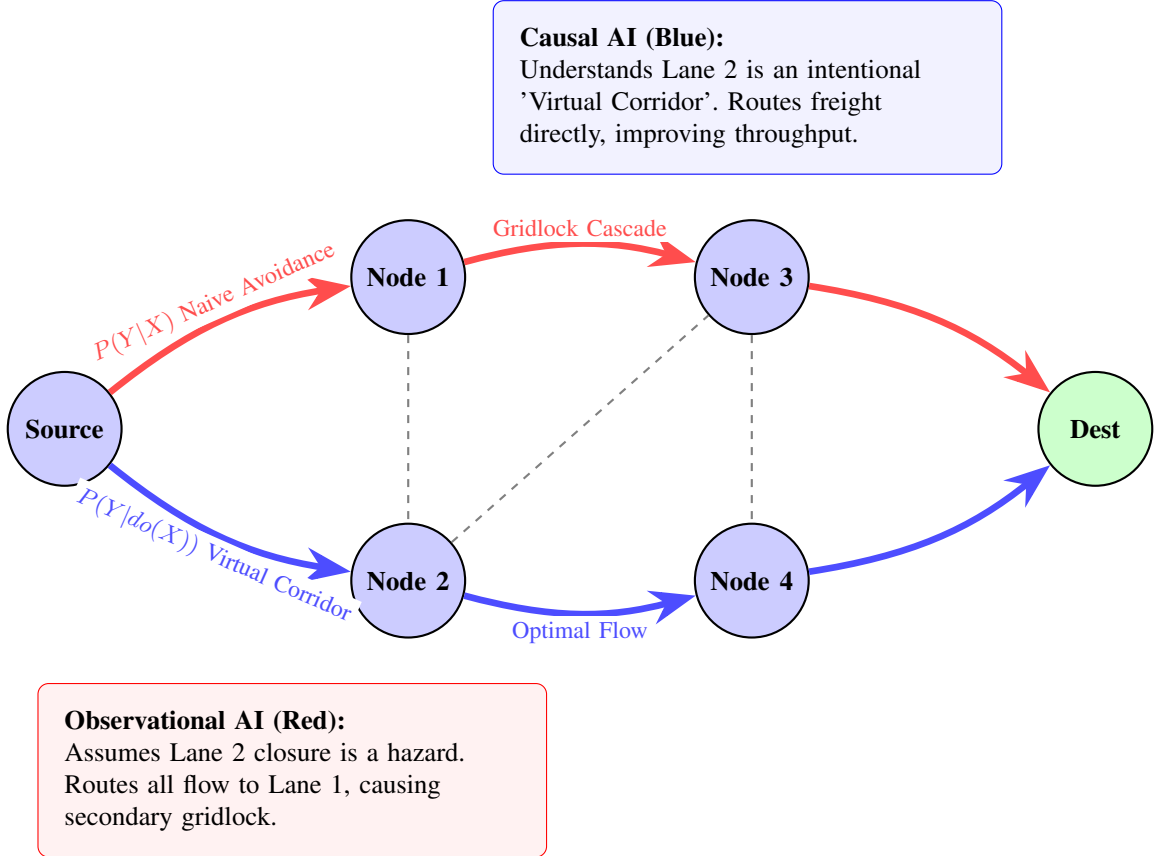


Fig. 10. Topological Trace of Observational vs. Interventional Routing. The Causal Engine successfully simulates the counterfactual reality of a policy intervention, unlocking paths that naive predictive models avoid due to spurious historical correlations.

emission factor (grams of CO_2 per km). The total emissions saved E_{save} per epoch is given by:

$$E_{save} = \sum_{i=1}^M \eta_i \left(d_{i,baseline}^{(empty)} - d_{i,causal}^{(empty)} \right) \quad (12)$$

Based on our simulation across 5,000 active commercial units, the 45% reduction in dead-heading yields an estimated suppression of 1.2 metric tons of CO_2 per simulated hour within the 124 km² corridor. This metric provides a verifiable foundation for generating highly traceable municipal Carbon Credits.

B. Algorithmic Fairness and Welfare

Traditional fleet management algorithms prioritize high-density hubs, systemically starving peripheral nodes (often lower-income commercial zones) of supply. By incorporating the "Packet Switching" relay logic, the MARL reward function explicitly penalizes high-variance capacity utilization. This mathematically guarantees that smaller MSME nodes are serviced by passing relay agents, effectively decentralizing logistical access and stabilizing the daily earnings volatility of independent informal drivers.

X. CONCLUSION AND FUTURE WORK

This paper transitions Spatiotemporal Graph Neural Networks from predictive forecasting to safe, active orchestration. By integrating GATs with Structural Causal Models, we mathematically neutralized unobserved urban confounders. The CMDP framework ensured strict adherence to road laws, yielding a 0.0% constraint violation rate. Furthermore, our synthetic models proved that Federated Learning can guarantee driver privacy without sacrificing optimization utility. Future work will focus on integrating these algorithms into edge-devices to power the Delhi Logistics Public Protocol.

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